

MATH Unit 3 Question Bank — Probability & Statistics

Mathematics (MATH) · Unit 3 · Probability & Statistics · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Explain Conditional Probability with a neat diagram and a suitable example.
2. Discuss the role of Bayes' Theorem in Probability & Statistics.
3. Explain Random Variables with a neat diagram and a suitable example.
4. Illustrate Binomial and Poisson Distributions with a worked example.
5. Discuss the role of Normal Distribution in Probability & Statistics.
6. Write short notes on Correlation and Regression.

Part B — Long answer (10 marks)

1. Define Conditional Probability. Derive or justify the key properties involved.
2. Explain Bayes' Theorem with a neat diagram and a suitable example.
3. Write short notes on Random Variables.
4. What is meant by Binomial and Poisson Distributions? Discuss its advantages and limitations.
5. Explain Normal Distribution with a neat diagram and a suitable example.
6. Define Correlation and Regression. Derive or justify the key properties involved.
7. Illustrate Conditional Probability with a worked example.
8. Explain Bayes' Theorem with a neat diagram and a suitable example.
9. What is meant by Random Variables? Discuss its advantages and limitations.
10. Define Binomial and Poisson Distributions. Derive or justify the key properties involved.
11. Discuss the role of Normal Distribution in Probability & Statistics.
12. Explain Correlation and Regression with a neat diagram and a suitable example.